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SurgeAI

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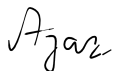
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Abstract

Advertising on traditional platforms such as TV, YouTube, and Facebook can be expensive and does not guarantee the ads are reaching an audience who actually is interested in them. This project, SurgeAI presents an AI-based web application that utilizes conversations from Reddit, Twitter, and Quora to discover consumers who really need their product. Leveraging Natural Language Processing, sentiment analysis and topic modeling, the platform validates product ideas, identifies user personas, and generates AI supported responses that establish connections between businesses and the right audience. SurgeAI will offer marketing that gets results, specifically designed for start-ups and small businesses. The project aims to provide an infinitely iterative and intelligent solution that makes market validation and advertising expenses lower.

Executive Summary

SurgeAI is designed to counter the limitations of traditional marketing analysis and content creation by offering a unique, automated platform powered by Artificial Intelligence AI and NLP. Traditional market trend analysis and promotional content development are resource-intensive, human centric, and quite inaccurate. Months of data gathering and data analysis from various channels are invested in it, limiting scalability and efficiency. These days, tools also lack depth in sentiment capture, topic relevance, and user interaction, causing weaker and broader marketing outcomes.

SurgeAI addresses these problems by providing an AI-based marketing automation platform that interprets social data, gains meaningful conclusions, and constructs optimized marketing comments automatically. The system integrates a multitude of technologies, including Reddit and Twitter APIs to gather information, Hugging Face Transformers to process NLP, and the OpenAI API or any other API to create smart comments. Automation minimizes human labor, streamlines campaign execution, and maximizes data-driven marketing.

SurgeAI follows a five-layer design model consisting of Presentation, API, Business Logic, Data, and External Integration layers. All the layers are built independently to facilitate modularity and scalability. Data is governed by the Data Layer through PostgreSQL and Redis for effective storage and caching, and the Business Logic Layer provides necessary functions like authentication, campaign management, and AI processing. The system is also safe for information with bcrypt password hashing, JWT authentication, and strict adherence to API rate limits and data privacy practices.

The system will be developed using the Agile Scrum approach to validate iterative development, regular testing, and continuous improvement. From user and campaign management, the system evolved to encompass web scraping automation, NLP processing, and AI-generated comment generation. Cost reduction was of utmost importance by utilizing open-source libraries and efficient caching mechanisms to reduce overhead costs and improve performance.

In Conclusion, SurgeAI offers a revolutionary and intelligent solution for automating marketing functions. With the integration of AI-powered analysis of data, real-time automation, and scalable infrastructure, it offers greater transparency, efficiency, and accuracy in digital marketing. The platform effectively closes the loop between human marketing and intelligent automation, offering a platform that is adaptable to evolving market requirements.

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Chapter 1 Introduction

TV ads, Facebook ads, YouTube commercial ads, and any form of online advertising have become very popular in the current corporate environment. Traditional advertising is very expensive, as well as does not include the assurance that it will reach the accurate needy audience. Also, people are not aware of what are the actual needs of people.

To overcome these issues, our project SurgeAI is an AI-based web platform that will get current chat from sites such as Reddit, Twitter, and Quora. Using NLP, sentiment classification, and topic modeling, the system will provide the need of solution or product among people and also the accurate needy persons. For advertising, the system will create AI-driven marketing statements to directly comment on those users. Small businesses and startups can show their ideas and compete competitively without the high-end price.

1.1 Purpose of this Document

The purpose of this document is to give the comprehensive technical details together with general description of our FYP, SurgeAI. The report comprises the description of our objective, design, implementation, testing, together with our limitations. The goal behind this project is to develop an AI-based web platform to validate the business idea and giving focused marketing intelligence by culling user chats from varied social networks. The project aims on minimizing advertisement expenditure and making it viable for startups to discover their potential audiences for their offerings.

1.2 Intended Audience

The target users for this report are the academic review panel, consisting of professors and examiners tasked to review this FYP. It is also good reading material for researchers and developers working on the field of AI, NLP, and sentiment analysis technologies. Business owners and startups will also find this report useful to know how one may utilize the power of AI for idea validation as well as affordable digital marketing.

1.3 Definitions, Acronyms, and Abbreviations

Following are all definitions, acronyms, and abbreviations used in this document.

FYP: Final Year Project

NLP: Natural Language Processing

ER: Entity Relationship

AI: Artificial Intelligence

TV: Television

API: Application Programming Interface

SaaS: Software as a service

SQL: Structured Query Language

BERT: Bidirectional Encoder Representations from Transformers

Sentiment Analysis: The task of classifying text as positive, negative, or neutral.

1.4 Conclusion

In conclusion, this report is structured to provide a comprehensive understanding of the project's development. It begins with the introduction which explain the problem of high cost advertising with idea validation and its solution as well. Then it explains the relevant work. The implementation phase of SurgeAI is then discussed, leading into an examination of the testing procedures, evaluation methods, and results obtained. The user manual to run this application is also explained in this report. Finally, the report concludes by summarizing the project's key outcomes and suggesting potential directions for future improvement and expansion.

Chapter 2 Project Vision

This chapter describes the full details of the system including the problem to be solved, problem elaboration, goals and objectives, project scope, Business Opportunity and stakeholders description.

2.1 Problem Domain Overview

The system SurgeAI is built to provide cost efficient solution to let people know about the company products and it also provides the feature for startups to first validate there idea. The system is AI-based web application that filters and analyzes the live conversations on social media platforms by scrapping or crawl like tools. Then passing these chats to NLP for Sentiment Analysis and Topic Modelling. The results of analysis will tell the company that how much your product is in demand by people. This will give idea to companies that wether they should proceed there idea or not. Also, the platform matches consumers who are interested in certain products or services and help companies to conduct targeted marketing. The system will provide AI-driven comments or replies to interact the targerted audience with prospects directly.

Thus it provides startups and small businesses with a cheap, scalable data driven market validation and advertisement possibility. The ultimate objective of this project is to cost efficient way of validation of idea and advertising and assist the companies, especially startup businesses to make awareness of there products among people.

2.2 Problem Statement

To identify that what people actually need is one of the biggest challenges when starting a business. Even after discovering a potential product or solution, companies often rely on inefficient marketing methods such as advertising on YouTube, TV, Google, Facebook, and other social media platforms without knowing whether the viewers actually need their product. This results in wasted marketing efforts and resources.

2.3 Problem Elaboration

The current techniques used to know what customers wants and address business product are facing major challenges that are related to related to cost, mediocre results. When new business ideas or start-ups created, the first challenge is trying to figure out what people really want. Even if they get to know the right product, they must have to spend a lot of money to advertise their product. There is also not a

direct method for a company to tell a person that in what problem you are facing, we have the solution. These problems are elaborated below.

2.3.1 Poor Idea Validation

For a business to become successful, a company must know what are the needs of the customers. Failing to know the customer needs will result in fall of the business. Often times, companies use assumptions, feelings or limited feedback to build products and services. Thinking that this will sell more and more. But the result is that countless startups never get their offerings meeting real customer needs, wasting time, energy and money. So, there must be an intelligent system to let startups know the demands of people.

2.3.2 Inefficient and Costly Advertising

Advertising through YouTube, TV, Facebook and other social media applications is very common these days but it takes a lot of money from companies to advertise their products. Once a company have found a product or solution, then it jump over the next hurdle that is marketing. Even the currently big companies that are although running successfully, spend a lot of money on marketing on these platforms. Moreover, the formats used to deliver ads now are designed to reach the most people possible, not necessarily the right people. That is just inefficient way of approaching people. Because companies have to pay lots of money advertising to people who might not even want the product. There is no guarantee that the audience are actually interested or searching for a solution like that, so that yields very low engagement and also conversion rates. Thus, there must be a system to cater this problem and especially for startups who cannot afford high-end advertising at the start.

2.3.3 Gap Between Customer and Company

In addition to these problems, there is not any intelligent system that can fill the gap of direct advertising to accurate needy persons who are actually the one who need the company specific product. A person cannot directly contact with the company, instead there is always a middle man who takes some commission. Our system aims to solve all this problem.

2.4 Goals and Objectives

The main objective of SurgeAI is to create an AI based web application that will help startups and small businesses to validate their product ideas as well as do targeted marketing, using real time discussions on multiple social platforms. It seeks to help digital marketing become more efficient and inexpensive.

The objectives of our project are:

1. To collect and analyze real-time discussions from Reddit, Twitter, and Quora using APIs and automated crawlers which will be filtered based on user specific keyword input.
2. To apply NLP and sentiment analysis techniques for understanding public opinion to check whether idea is in demand or not.
3. To identify target audiences and trending keywords relevant to business domains.
4. To generate AI-based marketing comments that will be posted by company to targeted persons.

2.5 Project Scope

This project is to build an AI based web application for startup and small business owners who want to validate their product ideas, and do the targeted marketing analysis by analyzing the real time discussions happening in social media platforms like Reddit, Twitter, Quora. The application will allow business to:

- Gather and analyze public conversations and posts from Reddit, Twitter and Quora based on the keyword they provide.
- Get a final opinion about the idea or product that whether should they launch and advertise their product by using NLP and sentiment analysis of posts.
- Create AI-powered marketing responses to posts where users express an interest in similar products or services.
- Provide an easy to use interface for performing actions easily

This project will be built with a combination of Next.js on the frontend, Python FastAPI will be used at the backend and NLP will be implemented in Python. The application will interact with a backend database through APIs. Database will be PostgreSQL. The development will be done with Agile Methodology and project management is maintained under SCRUM.

The project deliverables will include:

- A functional web-based application for businesses for idea validation and targeted marketing.
- A user manual for the application
- A documentation of the design, implementation, and testing process

The project scope does not include the following:

- Providing hosting or server infrastructure for the application

- Developing mobile applications (iOS or Android)

This project scope will be used as a guide throughout the development

2.6 Sustainable Development Goal (SDG)

The primary focus of this project is SDG-9: Industry, Innovation, and Infrastructure and its secondary focus is SDG-8: Decent Work and Economic Growth.

2.6.1 SDG-9 Industry, Innovation, and Infrastructure

SurgeAI drives innovation and great relief for startups around the globe by giving them access to intelligent AI-based marketing web application. It leverages AI and machine learning technology to develop smarter and more efficient business validation and marketing strategies. This promotes the growth of sustainable, technology based industries, spreading culture of innovation. The Figure 2.1 shows SDG-9.



Figure 2.1: SDG-9: Industry, Innovation and Infrastructure.

2.6.2 SDG-8 Decent Work and Economic Growth

SurgeAI accelerates economic growth. SurgeAI is an economically encouraging environment that fosters entrepreneurship by reducing marketing cost. The platform provides features that enable new digital marketing strategy to compete effectively, generating more sustainable opportunities for economic growth of a business. This is closely aligned with the aims of SDG 8, as shown in Figure 2.2, which promotes sustained, and sustainable economic growth, productive employment, and decent work for all.

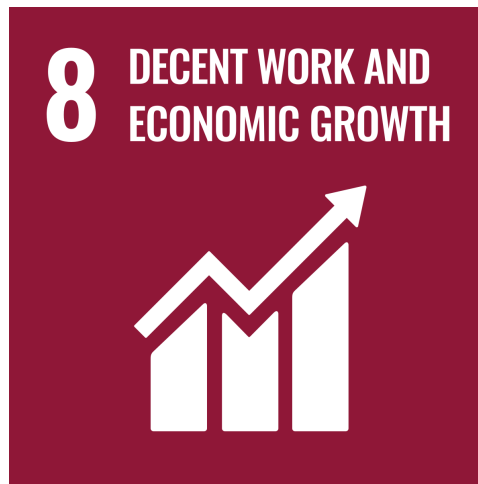


Figure 2.2: SDG-8: Decent Work and Economic Growth.

2.7 Constraints

The constraints that we will be facing in development of this project are:

- Getting chats from Reddit, Tweeter and Quaro is limited.
- These platforms APIs, especially of Tweeter are very difficult to get and they provide limited posts. We cannot get unlimited posts per account.
- The operational resources such as time to train a model and testing are also limited, because we are only three members working on this project and we only have time of two semesters.

2.8 Business Opportunity

SurgeAI provides an enormous business opportunity by allowing targeted and cost efficient marketing with idea validation for startups. Today, businesses pay enormous amount on blank ad campaigns that frequently do not reach the desired audiences. SurgeAI uses artificial intelligence to fill this blank by matching businesses with Reddit, Tweeter and Quaro users that are engaged in related conversations. The AI-based solution allows small, medium and even big sized businesses to reach potential customers and gain better visibility, save on marketing expenditures, and achieve better conversion rates.

In addition to this, with the data-driven approach increasingly adopted by digital marketing, SurgeAI has the potential to transform into a B2B SaaS platform, with subscription based offerings to marketing departments, startups, and digital agencies. This presents a scalable business model with potential for growth into other social media and online communities.

2.9 Stakeholders Description/ User Characteristics

The stakeholders of the SurgeAI system are individuals and organizations that will use this system. Business companies and administrators are the two main users of this system.

2.9.1 Stakeholders Summary

The business companies are small to medium enterprises, startups, and marketing teams seeking to promote products or services. They will use this system for validation of idea and for marketing of their product. They will input their idea and the system will provide them score of demand of their idea and further asks to advertise the product.

Administrators can see all the user activity and manage database.

2.9.2 Key High-Level Goals and Problems of Stakeholders

Startups, small businesses, as well as business users, want to target the correct users, save on the cost that is associated with advertising, and optimize conversions by targeting through the power of machine intelligence. Their biggest problem is the absence of affordable yet accurate targeting mechanisms for audiences combined with the need to depend on paid advertising that hardly reaches the actually interested users.

The administrators are concerned with delivering a stable and secure platform driven by AI. They are tasked with ensuring effective data collection and integrity of the system. Their key challenges are handling big sets of data in real time and coping with the platform APIs.

2.10 Conclusion

In conclusion, the vision of our project is to revolutionize the way businesses do marketing and audience targeting through the power of AI. SurgeAI aims to make the process of identifying potential customers more efficient, data driven, and cost effective. By providing startups and small businesses with real time insights and AI-generated outreach, our project holds the ability to replace traditional, expensive advertising methods with smarter and more personalized engagement.

Chapter 3 Literature Review / Related Work

This chapter contains details about the works done that relates to our project. It describe some of the applications that are somehow like our project. The primary objective of this chapter is to take a look at the previous related applications. This will give the reader to get some information about existing related work. Knowledge of the previous work can improve our system to cater the problems that existing works do not.

3.1 Definitions, Acronyms, and Abbreviations

FYP: Final Year Project

NLP: Natural Language Processing

AI: Artificial Intelligence

ROI: Return On Investment

CRM: Customer Relationship Management

Sentiment Analysis: The task of classifying text as positive, negative, or neutral.

3.2 Detailed Literature Review

This section describes detail of each work that relates to our project. Each related work describes summary of the work, critical analysis including strengths and weaknesses and its relation ship to our proposed work.

3.2.1 Validately and UserTesting

Current idea validation tools such as Validately and UserTesting [1] enable companies to collect user opinions about ideas and prototypes. But they are based mainly on highly controlled user testing and not natural discussion analysis. Such tools are fit for controlled testing but don't take into consideration the free and natural user opinions that are being expressed in public forums.

They have measurable outcome and formatted data but are not scalable nor automated and it might be biased. The opinions, suggestions, showed feelings, expressions and feedback collected from the client might be wrong because he or she might be testing or giving reviews only for money. They might do not have any interest for giving right feedback to the companies. Hence, they use paid test audiences, and that can bring along bias or restrict diversity of opinions.

This limit is overcome in SurgeAI, which uses natural user discussions found in places like Twitter,

Reddit and Quora to enable a more natural verification of user demand and market demand. This natural discussion that includes emotions helps to get proper idea validation and context marketing.

3.2.2 Hootsuite

Hootsuite [2] is a popular social media marketing and management platform which is used to schedule, post, and track content across many social media networks like Twitter, Facebook, Instagram and LinkedIn. It offers a beautiful and easy to use dashboard from which users can interact with their followers, see analytics, and control team workflows. Analytics tools of Hootsuite provide details of engagement levels, post reach, and demographics of audiences, which enable companies to refine their strategy for the web. It also supports tools to advertise on Facebook, Instagram and other social media platforms.

The strong point of Hootsuite lies in multi-platform integration, user-friendliness, and time-saving automation. It allows marketers to manage each social profile from a single location and thus enables consistent branding and participation. Although it is an amazing tool but is also not feasible when comes to advertising. It does not talk about conversations to determine user intent and sentiment. Like what company is going to advertise, is actually demanded by there followers?

SurgeAI extends the concept proposed by Hootsuite by including sentiment analysis and idea validation as a natural complement to the data pipeline. While Hootsuite is focused on scheduling and administration of engagement, SurgeAI attempts to decipher real-time demand in markets based on chats that happen over Reddit, Twitter, and Quora and responds intelligently by deploying AI-generated messages. Here, SurgeAI complements and extends the functionality of Hootsuite towards contextually smart marketing automation.

3.2.3 HubSpot

HubSpot [3] is an integrated marketing, sales, and CRM software built to enable companies to bring in visitors, convert visitors to leads, and close leads as customers via marketing automation. It has modules for email marketing, content creation, lead scoring, and campaign tracking. HubSpot's automation workflows help marketers deliver messages that are tailored to customer behavior and preference. Its integrated analytics dashboard makes it possible to track campaign effectiveness and ROI, and connections to social media and third-party services make it easier to use for digital marketing.

Another point of strength of HubSpot is its powerful automation and CRM integration, and it allows companies to maintain constant communication with leads and clients. It has a smooth lead to conversion experience. However, its automation depends a lot on highly structured customer information and

contact lists and not on free social discourse. It does not have any facilities to collect and mine real-time, unstructured information from public forums or social conversations. Lastly, as being a good inbound marketing system, its cost of subscription is too high for a young startup.

The new SurgeAI system finds its root in HubSpot automation workflows but improves upon them to add unstructured public data analysis. Instead of relying on closed-ended customer lists, SurgeAI discovers potential customers naturally based on sentiment and topic analysis of content and questions asked on Reddit, Twitter, and Quora. By making contextual AI comments to outreach, SurgeAI bridges the gap between automated marketing and AI-driven consumer discovery and makes intelligent marketing possible among startups at a fraction of the cost.

3.2.4 Brandwatch

Brandwatch [4] has been a well-known platform that offers real-time consumer insights by aggregating and analyzing data from a range of social networks, forums, and blogs. Brandwatch allows business organizations to track brand mentions, trends, and sentiment in the marketplace. Brandwatch uses NLP to determine opinions and discern emotional tone among posts. Brandwatch sentiment visualization and audience insights dashboards are used to enable marketing teams to make strategic choices based on conversations that are social.

Brandwatch's greatest strength is that it has the capacity to gather enormous social media data and uses sophisticated analytics to accurately treat it. It is customizable and has visualization that gives it a high value among large enterprises. Brandwatch, however, is a subscription product and its marketing has been aimed toward corporates, therefore economically out of reach for startups or small enterprises.

SurgeAI borrows a page out of Brandwatch's data gathering and sentiment monitoring pipeline but differs in focusing on product validation and marketing automation. While Brandwatch provides you with just the analytics, SurgeAI leverages AI to provide you with smart answers and product recommendations, making it a more proactive and cost effective option for startups.

3.2.5 BERT

BERT [5] is a transformer based NLP model that revolutionized text understanding tasks, including sentiment analysis, question answering, and topic modeling. BERT learns contextual representations of words in sentences, allowing it to capture nuanced meanings that traditional models often miss. It has since become the foundation for many sentiment analysis and text classification tools.

BERT provides best-in-class accuracy and transfer learning for NLP applications. Its computational cost, however, makes it less suitable for lightweight or real time application deployment that has not

been optimized. Although, it holds best text interpretation, it still needs to be fine tuned for datasets specific to a certain domain, which can be time consuming.

SurgeAI incorporates sentiment analysis models built on BERT to examine conversations throughout Reddit, Twitter, and Quora. We may adapt BERT to discuss data, SurgeAI can identify if users indicate needs, frustrations, or positive interest in a product. This directly aids the platform concept of validation and focused marketing objectives.

3.2.6 Meltwater

Meltwater [6] is a media intelligence and analytics platform that offers real time tracking of web news, blog posts, and social media content. It is mainly utilized for reputation tracking, competitor monitoring, and market trends. Meltwater uses NLP to determine trends, sentiment, and important influencers of conversations that happen online. Meltwater provides customizable dashboards that graphically represent media coverage and sentiment patterns over time, allowing companies to effectively follow their brand perception and market trends.

Meltwater's greatest strengths are its powerful data aggregation engine, real time alerting, and robust visualization. It does a good job to help organizations monitor their media track and derive insights regarding trends in the market. Its platform, though, focuses more on passive analysis than on active engagement. It has no automated responses and sentiment outcome based targeted marketing action mechanisms. It also carries a high tier enterprise level pricing that prevents it from being used in small business and startup settings that need affordable intelligence tools.

SurgeAI complements Meltwater's analysis capabilities and fuses media monitoring and AI marketing automation. Where Meltwater gives you insights, SurgeAI goes a step further and acts upon insights and creates AI-powered, contextually aware, personalized comments for prospective customers in real-time. With this, startups are able to not just see market sentiment but interact intelligently with users signaling interested needs and gain validation as well as marketing results from a single, unified system.

3.2.7 Buffer

Buffer [7] is a social media management suite that enables individuals to schedule, measure, and manage numerous social accounts from a single dashboard. It focuses on making content scheduling and tracking of performances easier for Twitter, Facebook, LinkedIn, and Instagram. Buffer also provides insights to help post reach, engagement rate, and audience growth to help individuals refine their social strategy.

The beauty of Buffer lies in its ease, cost effectiveness, and ease of use, and its best suited to startups and very small businesses. It primarily focuses, though, on scheduling and reporting, and it does not

have sophisticated tools like real time sentiment tracking or AI-based recommendations. Its reports are descriptive, not predictive, making it less valuable as a strategic market discovery resource.

SurgeAI improves the foundation of Buffre by adding AI and NLP capabilities. Instead of merely managing scheduled content, SurgeAI continuously monitors public forums and sentiment to identify potential customers and deliver contextually appropriate, AI-generated responses. SurgeAI thus offers a smarter, more aggressive solution to discovering and engaging audiences.

3.2.8 SEMrush

SEMrush [8] is a complete digital marketing and search engine optimization suite that provides keyword research, competitor tracking, backlink analysis, and content optimization. It helps organizations gain more web presence with high ranking search phrase discovery and insight into their competitor strategy. SEMrush also integrates ad and content marketing data, allowing users to track keyword trends and benchmark campaign success across many channels.

Semrush has a major strength in having a vast database and deep analysis, which make it extremely useful for keyword research and competitor analysis. It has reports that include data driven, actionable SEO metrics to facilitate data driven marketing decisions. Its main weakness, however, is that it is based purely on search and not on social or conversational data. It cannot measure real time audience sentiment or trends in conversations across social media, which makes it not very useful in finding out about customer conversations that are organic.

The new system, SurgeAI, goes a step further from SEMrush keyword and SEO analysis and emphasizes social media such as Reddit and others. SEMrush determines what users search for, while SurgeAI determines where and how they talk about products online. By monitoring user generated content via NLP and sentiment tracking, SurgeAI fills the gap between social intelligence and SEO insights and provides companies a direct means of touching the concerned audiences in real time conversations.

3.2.9 MonkeyLearn

MonkeyLearn [9] is a text analysis artificial intelligence platform that allows users to derive insights from unstructured text using machine learning models that are fully customizable. It also offers pre-trained tools for keyword extraction, sentiment analysis, and topic classification. MonkeyLearn is broadly used in the process of automating text workflows, for instance, analyzing customer reviews, social media comments, and customer feedback, to measure user sentiment and trends.

The convenience and versatility of this application makes it fit for a wide range of NLP-based applications. Its model builder, which is visual, makes the development of custom classifiers easy and allows

non-technical individuals to derive useful insights. However, in using user sourced datasets for training the models and being less scalable in the event of working with huge social media feeds in real time, it falls a little.

SurgeAI closely replicates MonkeyLearn NLP model techniques for sentiment analysis capabilities but applies them in a constantly evolving social environment like Reddit, Twitter and Quaro. SurgeAI examines real time conversation to find relevant discussions and determine their sentiment using NLP. This method translates NLP insights into actionable engagement for businesses, making social engagement data actionable in real time.

3.2.10 Talkwalker

Talkwalker [10] is a social listening and analytics tool that monitors social media, blog, forum, and news site discussions online. It uses NLP and image recognition techniques to identify brand mentions, emerging trends, and audience sentiment. Talkwalker allows marketing teams to see what people are saying and in what people are taking interests and discover influencers who want conversations about their product.

Strengths of Talkwalker include broad coverage of online sources and advanced AI-powered analytics that help to draw rich consumer insights. It offers visual dashboards, competitive benchmarking, and real time alerts for brand monitoring. However, it is more biased towards analytics and reporting rather than user personalization or AI-powered content creation. Its enterprise pricing scheme can also be restrictive for small companies and startups.

SurgeAI improves on Talkwalker concept by transitioning passive social listening into active engagement. Where Talkwalker analyzes trends and opinions, SurgeAI identifies specific users expressing needs or concerns that align with a company product and develops AI-based responses to such users. This makes SurgeAI a more task oriented, cost efficient solution with good specifications for startups that need direct and intelligent contact with audiences.

3.3 Literature Review Summary Table

The summary of related works relevant to our proposed project, SurgeAI, is presented in 3.1, highlighting key applications, their features, relevance to the project, and identified limitations.

Table 3.1: Summary of related works

Application	Features	Relevance	Limitations
UserTesting [1]	Human insights and usability testing platform	Provides structured feedback from real users; aligns with SurgeAI's goal of understanding user intent and improving targeting precision	Limited to invited participants, lacks real world spontaneous user data
Hootsuite [2]	Social media management and post scheduling across multiple platforms	Demonstrates automation in content delivery and engagement tracking, relevant to SurgeAI's goal of automating outreach	Focused on posting and analytics, not user discovery or context based targeting
HubSpot [3]	Inbound marketing, CRM, and email automation platform	Offers insights on customer journey automation that SurgeAI can apply for Reddit-based engagement	Primarily focused on owned audience databases, not external community insights
Brandwatch [4]	Consumer intelligence and social listening analytics	Highly relevant for sentiment analysis and trend detection key to understanding Reddit discussions in SurgeAI	Expensive and limited access to certain online communities like Reddit
BERT [5]	Deep bi-directional transformer for natural language understanding	Core NLP model that can enhance SurgeAI's ability to comprehend Reddit posts and detect contextual intent	Requires high computational resources and fine tuning for domain specificity
Meltwater [6]	Media monitoring and analytics for PR and social sentiment tracking	Useful for analyzing brand reputation, similar to SurgeAI's focus on sentiment mining for marketing relevance	Primarily used for brand monitoring, lacks community level granularity
Buffer [7]	Social media toolkit for content scheduling and analytics	Illustrates workflow simplification and automation models SurgeAI could adopt for marketing actions	Focused on post management, not insight extraction or audience discovery
SEMrush [8]	SEO, keyword research, and competitor analysis tool	Helps identify search intent and trends relevant for aligning SurgeAI's Reddit topic discovery with keyword intelligence	Concentrated on search engine data, not conversational or community contexts
Talkwalker [10]	Social listening and analytics across major social platforms	Valuable for real-time social trend monitoring, conceptually similar to SurgeAI's Reddit user analysis	Limited Reddit coverage and less focus on personalized targeting
MonkeyLearn [9]	AI text analysis for sentiment and topic classification	Can support SurgeAI in automating text classification and emotion detection in Reddit comments	Limited scalability and may require integration with larger NLP systems

3.4 Conclusion

In conclusion, the above sections explain current digital marketing and social listening platforms still remain severely limited. Although systems like Hootsuite, Buffer, and Brandwatch post automatically and offer analytics, they do not have intelligent, need based targeting by real time user conversations. On analysis of these systems, it is clear that a great opportunity lies ahead for improvement in directly linking businesses with interested audiences. SurgeAI fills this gap by providing an AI-driven, easy to use web application that integrates real time analysis of data, sentiment identification, and smart interactions enabling startups and small enterprises to test ideas and reach the right people in an efficient manner.

Chapter 4 Software Requirement Specifications

This is one of the major characteristics of our project. It contains functional and non-functional requirements, use cases, hardware and software requirements, and GUI + Database design.

4.1 List of Features

- System will allow registering and logging in.
- It will receive input of project ideas, as well as pain point filtering keywords.
- It will connect with Reddit API, X API and Google Trends API and scrap websites such as Quora to acquire user content.
- Our system will carry out pain point research and sentimental analysis of user information.
- System will present the potential clients to have reached according to sentimental analysis.
- It validates ideas through the analysis of the demand among the population regarding the pain point analysis and sentimental analysis.
- System will produce charts and graphs in order to visualize AI insights.
- It will produce a validation report that has been done with the detailed analysis.

4.2 Functional Requirements

The following are the general specifications of the project SurgeAI.

4.2.1 User Functional Requirements

- User can Register and Log-in into the platform.
- User can post a Validation idea.
- User can provide keywords to be used in pain point analysis.
- User will dashboard where Ai insights visualization is displayed.
- User will access dashboard of potential clients to make outreach.
- User can Export and download report of verified ideas.

4.2.2 Admin functional requirements

- Admin can login into the administration area.
- Admin can do performance monitoring and tracking of models.
- Admin can train and retrain classification models.

4.2.3 AI Engine Functional Requirements

- AI Engine will request data on such platforms as Reddit, X, Quora and Google Trends.
- The AI Engine deals with the text working with NLP models such as sentiment, intent, and topic classification.
- AI Engine identifies repetitive issues or problems among platforms.
- On analysis and competitor comparison, AI Engine creates a validation report.

4.3 Quality Attributes

- The system will be user friendly to the new users, providing high Usability.
- The platform will be receptive to the addition of additional sources of data such as APIs, providing scalability.
- The site will provide security of user credentials.

4.4 Non-Functional Requirements

- The system shall provide 99% uptime in our system.
- The system shall ensure credentials and authentication based on JWT and authentication based on encryption with SSL.
- The system will capable of producing a validation report in a period of 10-15 seconds.

4.5 Assumptions

- Internet connection should be available to user.
- APIs such as Reddit, X and Google Trends are stable.

4.6 Use Cases

Table 4.1: User Signup Use Case

Name		User Signup	
Actors		New User	
Summary		To create a new account, the user enters the necessary information such as name, email, and password. After account creation, the credentials are saved, and the user proceeds to log in.	
Pre-Conditions		Before registration, the email of the user should not have been used previously.	
Post-Conditions		The registration of the user is completed, and the user is notified about the registration through email.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	The user clicks on the “Sign Up” section on the home page.	2	The system opens the registration form.
3	The user enters their name, email, and password, and confirms the password.	4	The system checks all input fields, verifying email and password formatting.
5	The user submits the signup form.	6	The system confirms that the email is not already registered, stores the information, and creates a new user profile.
7	A confirmation message is displayed on the system.	8	The user is redirected to the login section or automatically logged in.
Alternative Flow			
9	The user provides an email that is already registered.	8-A	The system displays an error message stating that the email is already in use and prompts the user to use another one.
10	The user enters incomplete or invalid information.	8-B	The system highlights the incorrect fields and requests the user to correct them.
11	A technical error occurs during registration.	8-C	The system displays an error message and instructs the user to try again later.

Table 4.2: User Login Use Case

Name	User Login		
Actors	Registered User		
Summary	One logs in to the system by entering legitimate logins. Once the authentication is successful, access is given and the user redirected to his/her dashboard.		
Pre-Conditions	This requires that the user should already have an active account and not be in the process of being logged in.		
Post-Conditions	The user is authenticated successfully and sent to the main dashboard.		
Special Requirements	This process does not need any special hardware or software.		
Basic Flow			
Actor Action		System Response	
1	The user opens the login page.	2	The system presents the interface of logging in.
3	The user types out a valid email and password.	4	The system authenticates the given credentials and in case of a match, logs the user into the system to land him or her on the dashboard.
Alternative Flow			
5	The customer enters a wrong e-mail address or password.	4-A	The system denies the logging in process and gives an error message that requests the user to re-enter.

Table 4.3: Idea Submission Use Case

Name		Idea Submission	
Actors		User	
Summary		An authenticated user makes a new idea to be validated after filling in a form with the title of the idea, description, and other keywords related to the idea made. Then AI-based analysis is started	
Pre-Conditions		The system requires the user to log into the system then access submission form.	
Post-Conditions		The idea provided is entered into the system successfully and an automated analysis starts to analyze and collect correlated data online.	
Special Requirements		No extra hardware or software is required in this function.	
Basic Flow			
Actor Action		System Response	
1	The user clicks on the section “Submit Idea” of the dashboard.	2	The system shows the form of idea submission.
3	The user provides in the title, description, and the keywords associated with it.	4	The system verifies the appropriateness of each field.
5	The user submits the form.	6	The system records the idea and begins the data collection and analysis via AI.
7	The user submits the form.	8	An acknowledgment message appears, and this means that the idea has become a successful submission.
Alternative Flow			
9	The user provides incomplete or invalid information.	8-A	The system brings out the wrong or missing fields, and the user is called upon to correct the mistakes.

Table 4.4: Sentiment and Pain-Point Analysis Use Case

Name	Sentiment and Pain-Point Analysis		
Actors	AI Engine		
Summary	The AI engine processes the gathered online information concerning the idea given and provides insight into sentiment, trends, and repetitive pain points.		
Pre-Conditions	The concept needs to be effectively rolled out, and information sources (Reddit, Twitter, etc.) should be present.		
Post-Conditions	The data extracted such as sentiment scores and pain points are kept so that they can produce reports.		
Special Requirements	Constant API connection with data sources.		
Basic Flow			
Action of Actor		Response of System	
1	The idea keywords are passed to AI Engine.	2	The system retrieves information from integrated APIs (Reddit, Twitter, Quora, Google Trends).
3	The AI model assesses the mood, identifies keywords, and finds pain points.	4	These are stored in the database where they are to be processed.
Alternative Flow			
Action of Actor		Response of System	
5	Exits API request unsuccessfully or gives incomplete data.	4-A	The system re-tries the data retrieval or marks the request as failed.

Table 4.5: Admin Management Use Case

Name	Admin Management		
Actors	Admin		
Summary	The administrator controls and directs the back-end functions of the system such as account and user information and AI settings. The administrator has the ability to remove data connections and ensure overall stability of the system.		
Pre-Conditions	The administrator should be logged in using valid credentials and possess high-level access permissions.		
Post-Conditions	Any approved changes made by the administrator are stored in the database and mirrored throughout the system.		
Special Requirements	The admin must have the required permissions and a stable network connection to the server or database.		
Basic Flow			
Action of Actor		Response of System	
1	Using valid credentials, the administrator logs into the administration dashboard.	2	The system logs in the administrator and provides access to management features.
3	The administrator performs actions such as user management altering data sources, or updating AI settings.	4	The system processes these changes, updates the database, and confirms successful completion.
Alternative Flow			
Action of Actor		Response of System	
5	The administrator provides invalid input or performs an unauthorized action.	4-A	The system denies the operation, logs the error, and displays an appropriate warning notification.

4.7 Hardware and Software Requirements

This section outlines the minimum hardware and software specifications required to effectively develop, deploy, and operate the Idea SurgeAI system.

4.7.1 Hardware Requirements

- A good processor, preferably Intel Core i5 or higher.
- At least 8 GB of RAM required in order to smooth process and do multitasking
- A stable internet connection to enable consistent data collection from multiple platforms.
- A Solid State Drive (SSD) for faster system performance and improved data handling speed.

4.7.2 Software Requirements

- Backend framework of Python with FastAPI or Django is required to develop and implement RESTful APIs.
- Next.js is required to build a responsive and beautiful user interface.
- PostgreSQL datanbase is required for efficient database storage and access management.
- AI and data libraries such as transformers (like Hugging Face), Scikit-learn, Pandas, Tweepy, and PRAW are required for natural language processing, data processing, and social media data access.
- Visual Studio Code and GitHub are essential tools for efficient and centralized software development

4.8 Graphical User Interface

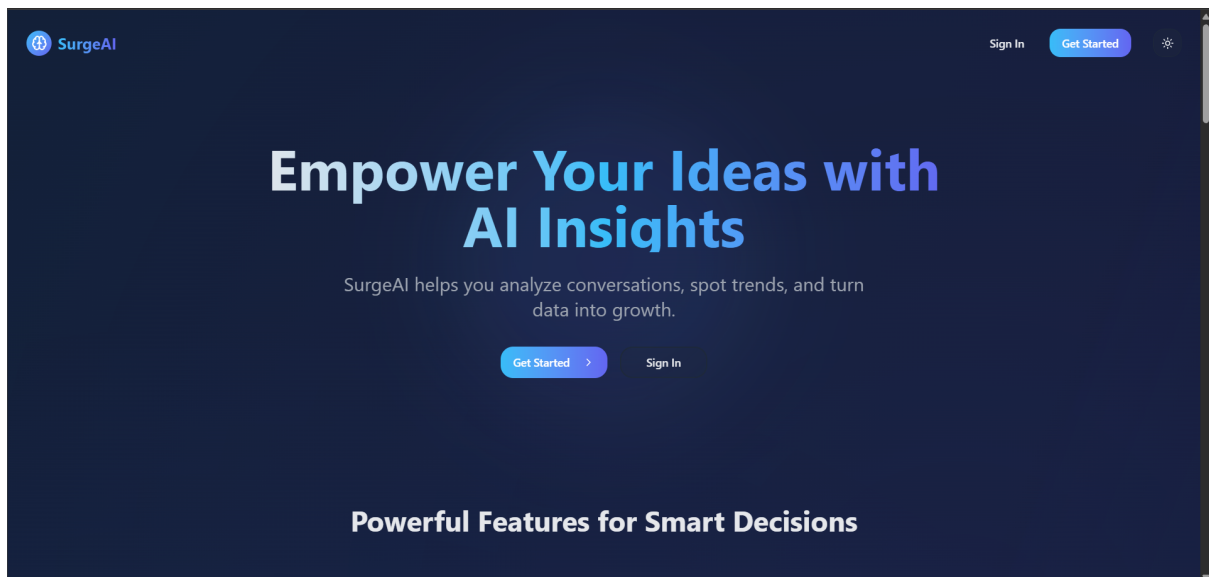


Figure 4.1: Landing Page

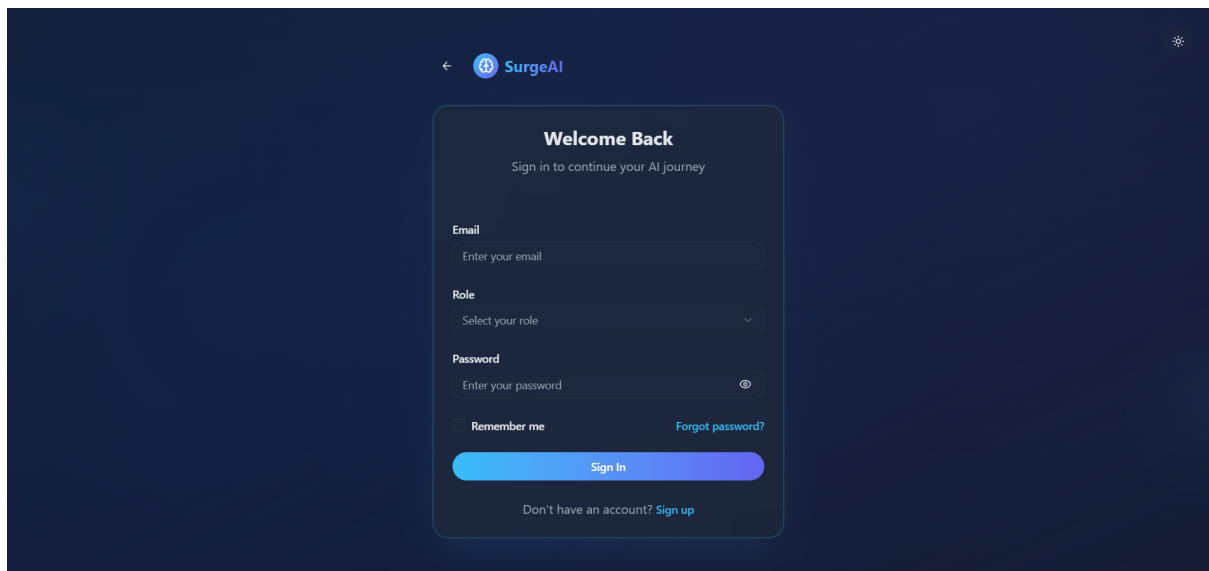


Figure 4.2: SignIn Page

The screenshot shows the first step of the SurgeAI account creation process. At the top, there is a back arrow, the SurgeAI logo, and a progress bar indicating 'Step 1 of 2' with a 50% completion mark. The main heading is 'Create Account', followed by the subtext 'Join thousands of users leveraging AI insights'. Below this, there are two input fields: 'Full Name' with the placeholder 'Enter your full name' and 'Email' with the placeholder 'Enter your email'. A blue 'Continue' button is positioned below the email field. At the bottom, there is a link that says 'Already have an account? Sign in'.

Figure 4.3: SignUp Page Step 1

The screenshot shows the second step of the SurgeAI account creation process. At the top, there is a back arrow, the SurgeAI logo, and a progress bar indicating 'Step 2 of 2' with a 100% completion mark. The main heading is 'Secure Your Account', followed by the subtext 'Set up your password and security'. Below this, there are two input fields: 'Password' with the placeholder 'Create a strong password' and 'Confirm Password' with the placeholder 'Confirm your password'. Both fields have an eye icon to toggle password visibility. A blue 'Create Account' button is positioned below the confirm password field. At the bottom, there is a link that says 'Already have an account? Sign in'.

Figure 4.4: SignUp Page Step 2

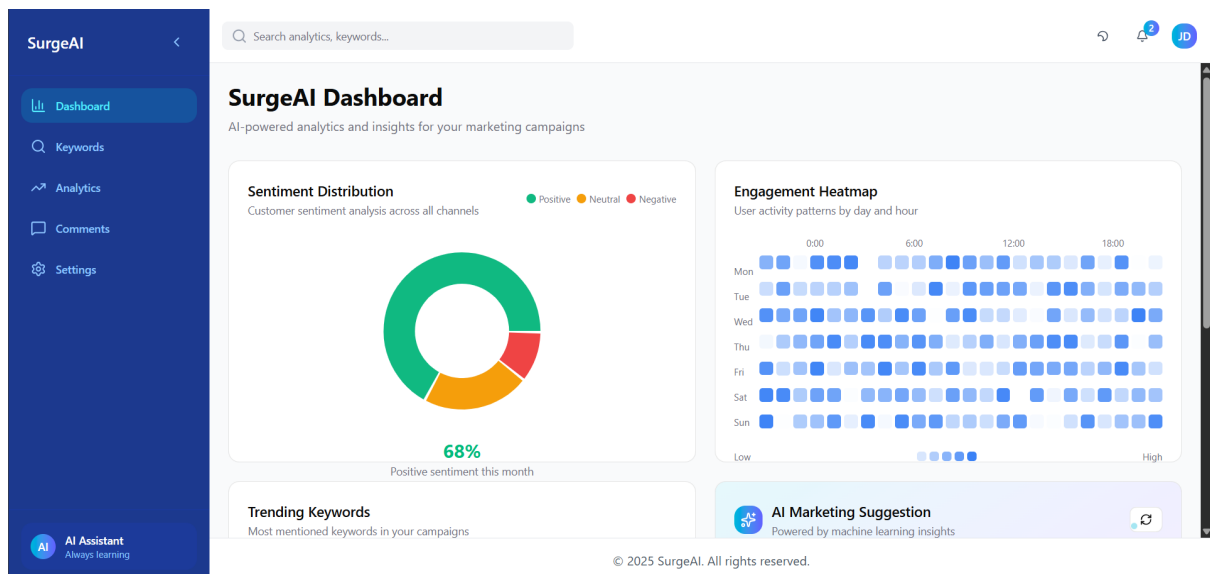


Figure 4.5: Client Dashboard

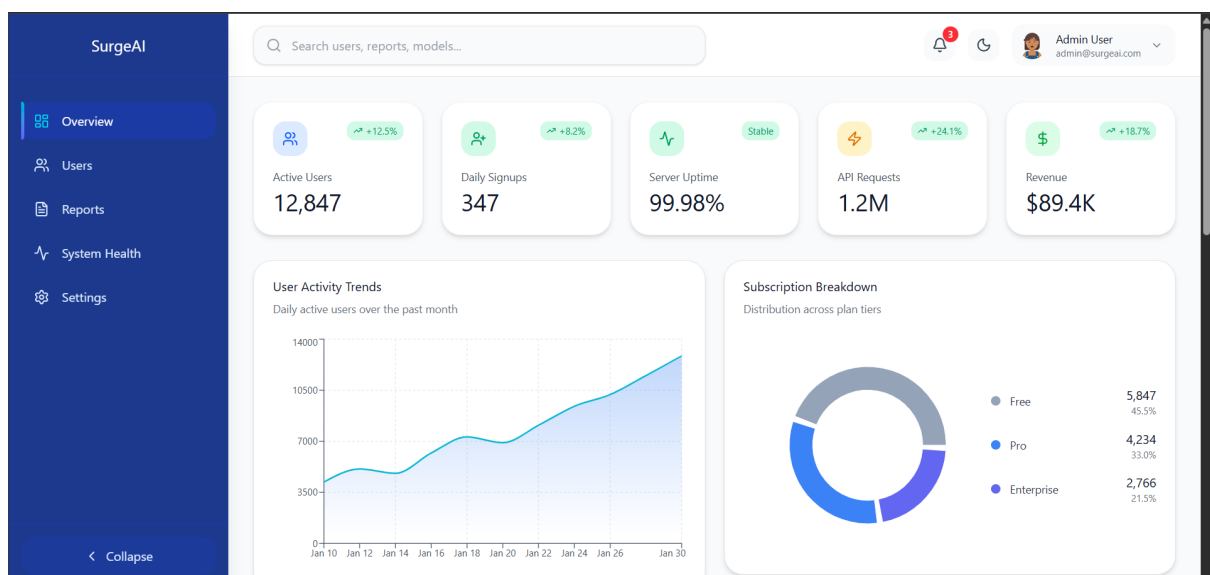


Figure 4.6: Admin Dashboard

4.9 Database Design

4.9.1 ER Diagram

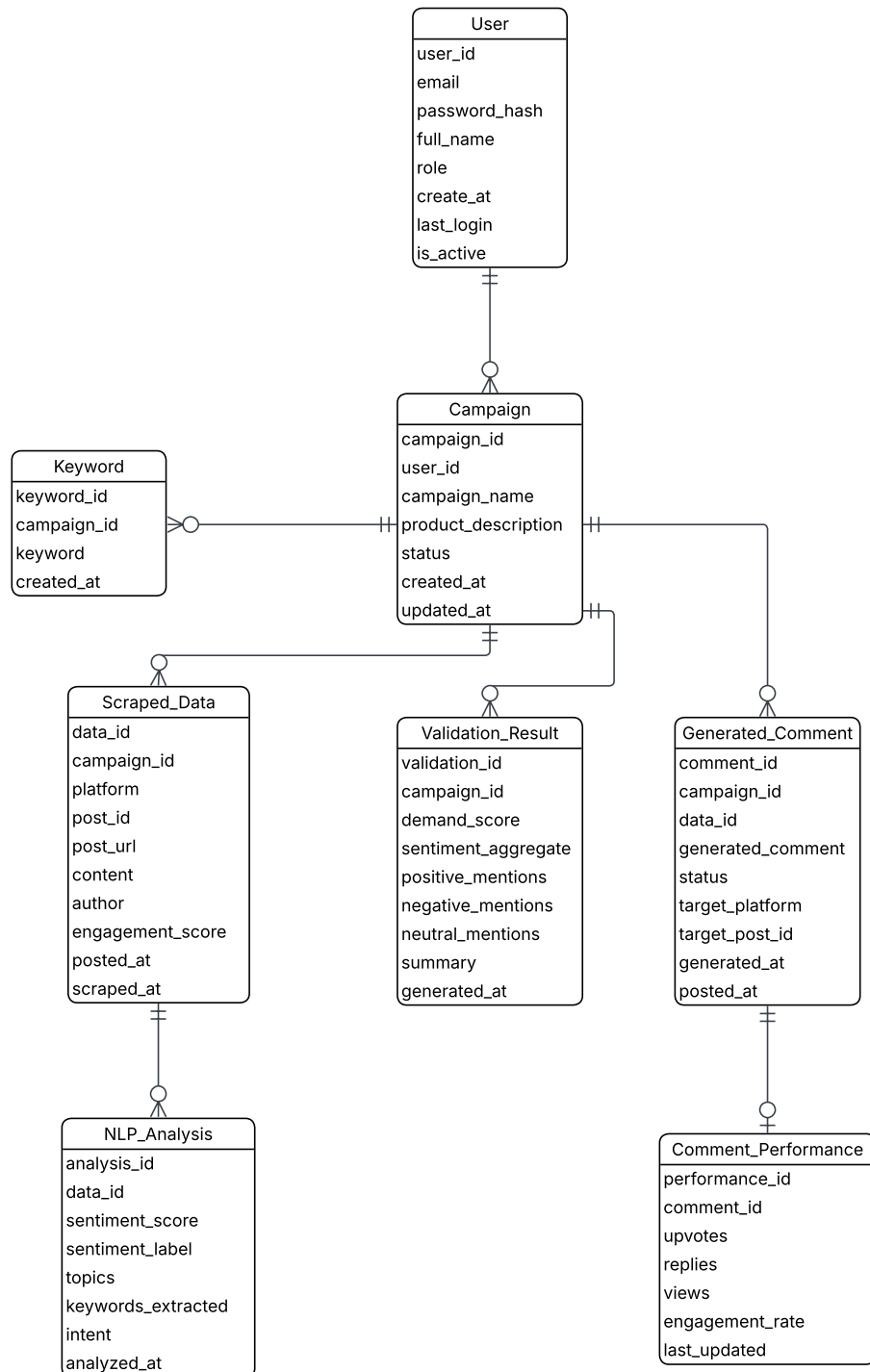


Figure 4.7: ER Diagram of SurgeAI

4.9.2 Data Dictionary

Table 4.6: Comprehensive Data Dictionary for SurgeAI

Entity	Attribute	Data Type	Constraints	Description
USER	user_id	INTEGER	PRIMARY KEY	Unique user identifier
	email	VARCHAR(255)	UNIQUE, NOT NULL	Login email
	password_hash	VARCHAR(255)	NOT NULL	Hashed password (bcrypt)
	full_name	VARCHAR(100)	NOT NULL	User's full name
	role	ENUM('user', 'admin')	DEFAULT 'user'	Access role
	created_at	TIMESTAMP	DEFAULT NOW()	Registration date
	last_login	TIMESTAMP	NULL	Last login timestamp
	is_active	BOOLEAN	DEFAULT TRUE	Account status
CAMPAIGN	campaign_id	INTEGER	PRIMARY KEY	Unique campaign identifier
	user_id	INTEGER	FOREIGN KEY → USERS	Campaign owner
	campaign_name	VARCHAR(268)	NOT NULL	Campaign title
	product-description	TEXT	NOT NULL	Product/idea description
	status	ENUM('active', 'paused', 'completed')	DEFAULT 'active'	Campaign status

Entity	Attribute	Data Type	Constraints	Description
	created_at	TIMESTAMP	DEFAULT NOW()	Creation date
	updated_at	TIMESTAMP	ON UPDATE NOW()	Last update
KEYWORD	keyword_id	INTEGER	PRIMARY KEY	Unique keyword identifier
	campaign_id	INTEGER	FOREIGN KEY → CAM- PAIGNS	Parent campaign
	keyword	VARCHAR(100)	NOT NULL	Search term
	created_at	TIMESTAMP	DEFAULT NOW()	Added date
SCRAPED _DATA	data_id	INTEGER	PRIMARY KEY	Unique data identifier
	campaign_id	INTEGER	FOREIGN KEY → CAM- PAIGNS	Parent campaign
	platform	ENUM('reddit', 'twitter', 'quora')	NOT NULL	Source platform
	post_id	VARCHAR(100)	UNIQUE, NOT NULL	Platform post ID
	post_url	VARCHAR(500)	NOT NULL	Direct post link
	content	TEXT	NOT NULL	Post text content
	author	VARCHAR(100)	NULL	Post author

Entity	Attribute	Data Type	Constraints	Description
	engagement_score	INTEGER	DEFAULT 0	Likes/upvotes count
	posted_at	TIMESTAMP	NULL	Original post date
	scraped_at	TIMESTAMP	DEFAULT NOW()	Collection date
NLP _ANALYSIS	analysis_id	INTEGER	PRIMARY KEY	Unique analysis identifier
	data_id	INTEGER	FOREIGN KEY → SCRAPED- DATA	Analyzed post
	sentiment_score	FLOAT	CHECK (-1 to 1)	Sentiment polarity
	sentiment_label	ENUM('positive', 'negative', 'neutral')	NOT NULL	Sentiment category
	topics	JSON	NULL	Extracted topics
	keywords- extracted	JSON	NULL	Key phrases
	intent	VARCHAR(50)	NULL	Detected intent
	analyzed_at	TIMESTAMP	DEFAULT NOW()	Analysis date

Entity	Attribute	Data Type	Constraints	Description
VALIDATION_RESULT	validation_id	INTEGER	PRIMARY KEY	Unique validation identifier
	campaign_id	INTEGER	FOREIGN KEY → CAMPAIGNS	Parent campaign
	demand_score	FLOAT	CHECK (0 to 100)	Overall demand score
	sentiment-aggregate	FLOAT	CHECK (-1 to 1)	Average sentiment
	positive_mentions	INTEGER	DEFAULT 0	Positive posts count
	negative_mentions	INTEGER	DEFAULT 0	Negative posts count
	neutral_mentions	INTEGER	DEFAULT 0	Neutral posts count
	summary	TEXT	NULL	AI-generated summary
	generated_at	TIMESTAMP	DEFAULT NOW()	Generation date
GENERATED_COMMENT	comment_id	INTEGER	PRIMARY KEY	Unique comment identifier
	campaign_id	INTEGER	FOREIGN KEY → CAMPAIGNS	Parent campaign
	data_id	INTEGER	FOREIGN KEY → SCRAPED-DATA	Target post

Entity	Attribute	Data Type	Constraints	Description
	generated-comment	TEXT	NOT NULL	AI comment text
	status	ENUM('draft', 'approved', 'posted', 'rejected')	DEFAULT 'draft'	Review status
	target_platform	VARCHAR(200)	NOT NULL	Posting platform
	target_post_id	VARCHAR(100)	NOT NULL	Post to reply to
	generated_at	TIMESTAMP	DEFAULT NOW()	Generation date
	posted_at	TIMESTAMP	NULL	Posting date
COMMENT _PERFORM- ANCE	performance_id	INTEGER	PRIMARY KEY	Unique performance identifier
	comment_id	INTEGER	FOREIGN KEY → GENERATED-COMMENTS	Tracked comment
	upvotes	INTEGER	DEFAULT 0	Upvotes/likes received
	replies	INTEGER	DEFAULT 0	Reply count
	views	INTEGER	DEFAULT 0	View count
	engagement_rate	FLOAT	DEFAULT 0.0	Calculated rate
	last_updated	TIMESTAMP	NULL	Last updation date

4.10 Risk Analysis

Our project contain technical and operational risks. These risks are elaborated below.

4.10.1 Technical Risks

Rate limits or access restrictions on Reddit, Twitter, and Quora APIs may affect data collection. Very limited amount of posts we will be able to get from these platforms. We may use scrapers or crawlers to get data, but the data collected by using these techniques gives so much unorganized data. So, the sentimental analysis will might not be so much accurate.

4.10.2 Operational Risks

At our current knowledge level, we lack of expertise in NLP or data engineering including its integration with project. Therefore, first we will have to learn these new technologies. About how to use and integrate with in our project. This could delay development.

4.11 Conclusion

In conclusion, this chapter explains the Software Requirement Specifications for the SurgeAI system, covering both functional and non-functional requirements required for its implementation. It introduces database design components, such as the ER diagram and a data dictionary defining entities, relationships, and constraints to maintain data integrity and system consistency. Also, the risk analysis section outlined possible technical and operational risks, including API constraints and no capability in NLP and data integration, that might affect the progress of development. Overall, this chapter gives a good foundation for system design and implementation by explicitly defining the data structures and possible challenges.

Chapter 5 High-Level and Low-Level Design

This chapter gives the low-level and high-level design of the SurgeAI system, explaining how its parts collaborate in order to provide scalability, efficiency, and maintainability. It specifies the development principles, subsystem design, and architectural framework that dictate implementation. The chapter also identifies the design principles, technologies, and frameworks employed to provide secure and modular system operations.

5.1 System Overview

The design of SurgeAI represents a five-layer architecture which promotes modularity and ensures separation of concerns. The Presentation Layer, powered by NextJS, is responsible for user interaction and rendering the web interface. The API layer (FastAPI) contains RESTful endpoints that manage authentication, authorisation and business logic. The Business Logic Layer consists of eight core subsystems, which are as follows: Campaign Management, Data Scraping, NLP Processing, Idea Validation, Comment Generation, and Analytics. The Background Processing Layer (Celery and Redis) is responsible for the execution of time-intensive tasks asynchronously, such as web scraping and NLP analysis. The Data Layer manages persistent storage and caching. PostgreSQL is used for persistent storage, while Redis is used for caching.

There is a predefined flow to the tasks running in the backend that is the API layer and the Business Logic layer combined. In other words, service-oriented design is followed, where each subsystem is modularised into a service with a well-defined interface.

The key design patterns that are used are the Strategy Pattern, Repository Pattern, and the Observer Pattern. Strategy pattern is used to cater for multiple platform scrapers, repository pattern is used to separate the business logic from the data access logic, and observer pattern is used for real-time status updates. The main focus of this architecture is horizontal scaling, which means that if more users or scraping tasks come in, just more server instances need to be added instead of upgrading the server. It essentially means that the system is stateless; each server can do the same job independently.

External integrations are done using adapter interfaces. Small connector classes, which are called adapters, handle the connection to the external APIs instead of directly connecting to them.

5.2 Design Considerations

5.2.1 Assumptions and Dependencies

5.2.1.1 Technical Assumptions

- Reddit and Twitter maintain stable APIs with reasonable rate limits.
- Quora lacks an official API so we'll be dependent on scraping via Selenium, which assumes that the platform's HTML structure remains stable.
- Reliable, high-speed internet connectivity for both the hosting infrastructure and the end user.
- Stable Python 3.9+ environment with the support of async/await operations in the backend.
- Stable NodeJS 16+ for NextJS execution.
- PostgreSQL is assumed to handle concurrent read and write operations efficiently.
- The response times relating to complex queries involving joins across multiple tables are to be acceptable; typically less than 500ms.

5.2.1.2 Operational Assumptions

- It is assumed that the user reviews and approves the AI-generated marketing comment instead of blindly posting it.
- It is expected that the platform scrapers will encounter English language content.
- Users are expected to respect the terms of service of the scraping platforms.
- At least one admin is assumed to be available.

5.2.2 External Dependencies

5.2.2.1 Third Party APIs

- Reddit API (PRAW)
- Twitter API v2
- OpenAI API

5.2.2.2 Open Source Libraries

- HuggingFace Transformers

- Celery
- SQLAlchemy

5.2.2.3 Infrastructure Dependencies

- PostgreSQL Database
- Redis Server
- Selenium WebDriver

5.2.3 General Constraints

5.2.3.1 Hardware and Software Environment Constraints

- System requires Python 3.9+, Node.js 16+, PostgreSQL 13+, and Redis 6+ with minimum 4 GB ram and 2 cpu cores.
- To automate the copying of data from Quora, our program needs to run a real web browser in the background so the server should be powerful enough and configured to allow a browser to be installed and run.

5.2.3.2 External API Rate Limits

- Reddit API allows 60 requests per minute.
- Twitter API provides 500,000 tweets per month on free tier.
- OpenAI charges per token.
- These limits have a direct impact on the campaign processing speeds and the overall operational cost.

5.2.3.3 Performance Requirements

- API responses must complete within 2 seconds.
- Background scraping must complete within 10 minutes per campaign.
- NLP analysis should be completed within 500 ms per post.
- The system should be capable of supporting 100+ users at a time

5.2.3.4 Security and Compliance Constraint

- bcrypt hashing with 12+ salt rounds should be used for passwords.
- JWT tokens expire after 24 hours.
- API keys must be stored in environment variables.
- Scraping should respect rate limits and must comply with the platform's terms of service.

5.2.3.5 Data Storage Constraints

- PostgreSQL is capped at 50GB and to tackle this data older than 6 months is archived.
- Redis cache is limited to 2GB with LRU policy.

5.2.4 Goals and Guidelines

5.2.4.1 Modularity and Separation of Concerns

- Each subsystem operates independently with well-defined interfaces.
- Service-oriented design is followed, and as different services are isolated from each other, it is easy to upgrade a single service without having any effect on the others.

5.2.4.2 Asynchronous Processing for Responsiveness

- User experience is preferred over immediate completeness.
- Time-intensive tasks will continue in the background while keeping API responses fast.
- Due to this, the user experience will be seamless, without any blocking delays.

5.2.4.3 Fail-safe and Graceful Degradation

- If non-critical components fail, the system will continue to operate.
- If, for some reason, a platform's API becomes unavailable, scraping would continue with the remaining platforms.
- If NLP analysis fails for some posts, then the analysis will continue on the remaining available posts.

5.2.4.4 Cost Effectiveness

- Operational costs for the users are minimised by favouring open source tools and free tier APIs wherever possible.
- Redis is used for caching and queueing, and efficient caching reduces redundant API calls.

5.2.4.5 Scalability Through Backend Workers

- Horizontal scaling is supported for computationally heavy tasks such as scraping and NLP analysis.
- This means that more celery workers can be added without code changes.

5.2.4.6 Security by Design

- Authentication and Authorisation are taken care of at the API layer.
- Role-based access prevents unauthorised access.
- Sensitive data follows best security practices.

5.2.4.7 Simplicity and Maintainability

- Simplicity is prioritised over complexity.
- The system is divided into layers with straightforward APIs and standard database design.
- Overengineering is avoided to ensure that the codebase is understandable.

5.2.5 Development Methods

The system is developed using Agile methodology, following the Scrum framework. Functional pieces are released one at a time in manageable steps. The project starts with core features like user management and campaign management, which are followed by data scraping, NLP analysis, and comment generation in the subsequent phases. Domain-driven design concepts are followed, according to which code is organised around business tasks such as scraping and NLP analysis instead of technical layers. The business logic is separated from infrastructure concerns, which isolates the core business domain from frameworks and databases. Apart from this, prototyping of NLP components is done to explore different pre-trained models and choose the one with the best accuracy and suitability.

5.3 System Architecture

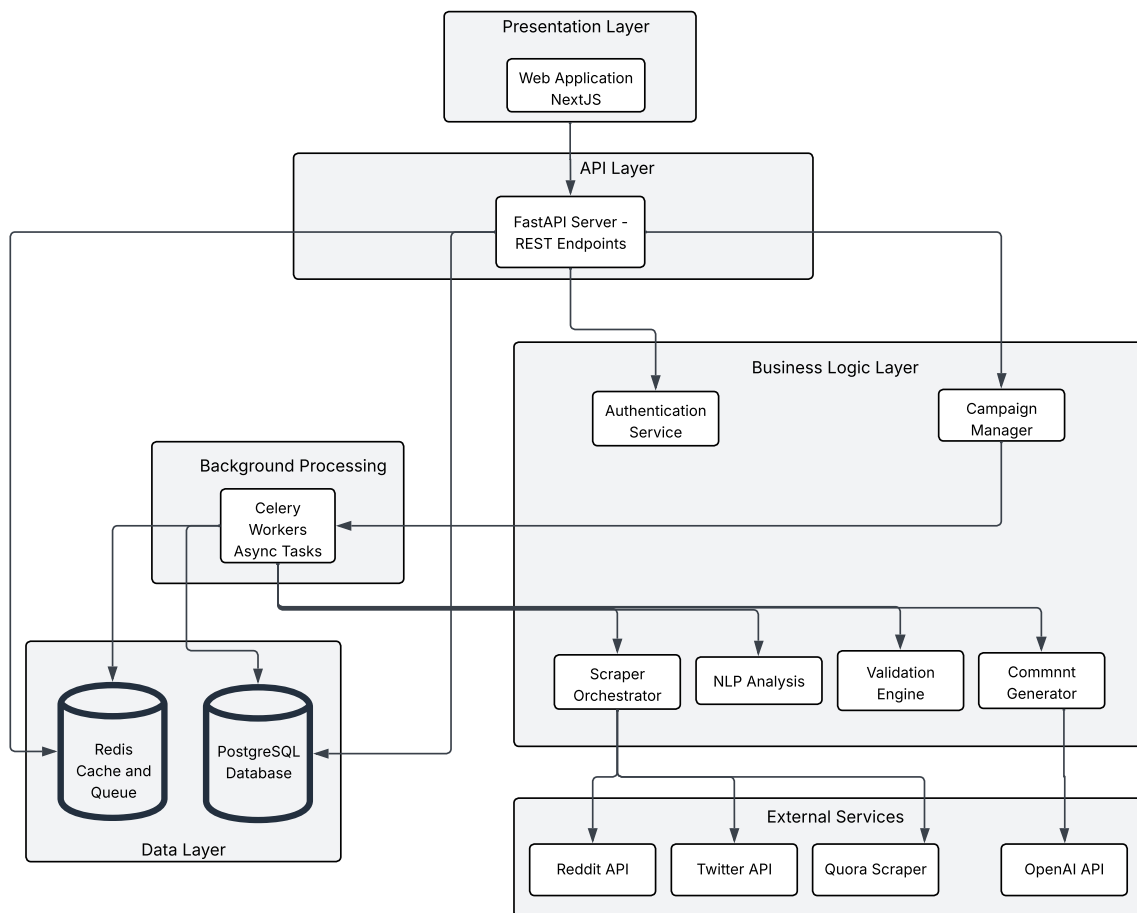


Figure 5.1: System Architecture of SurgeAI

It is a five-layer system divided into subsystems to isolate every service from each other. This leads to a well-defined structure where the role of every service is clearly defined. The Presentation Layer deals with all kinds of user interaction and provides the user request to the API layer. The API layer exposes all of the endpoints and is connected to two layers, namely the Data Layer and the Business Logic layer. The data layer provides the API layer with relevant data that is requested by a specific API. Redis provides cached data, and if that data is not available, then data is queried from the PostgreSQL database. The API layer also connects to two components of the Business Logic Layer, the Authentication service and Campaign management. The authentication service is responsible for, as the name suggests, authenticating the user. The campaign management service, upon creation of a campaign, triggers the celery workers, which in turn trigger the analysis pipeline. The first job in the pipeline is data scraping from relevant platforms, which are part of the external layer and connect to the scraper service. Once the data is scraped, then comes the second job, which is NLP analysis. NLP analysis performs senti-

ment analysis, topic modelling, and keyword extraction on scraped data using pre-trained models from Hugging Face transformers. After this comes the third job, the validation engine, which essentially uses the results of NLP analysis to produce and calculate demand scores, sentiment distributions and trend analysis. The fifth and the last job is comment generation, which gets context data from the previous jobs and uses OpenAI API to generate a marketing comment.

5.3.1 Subsystem Architecture

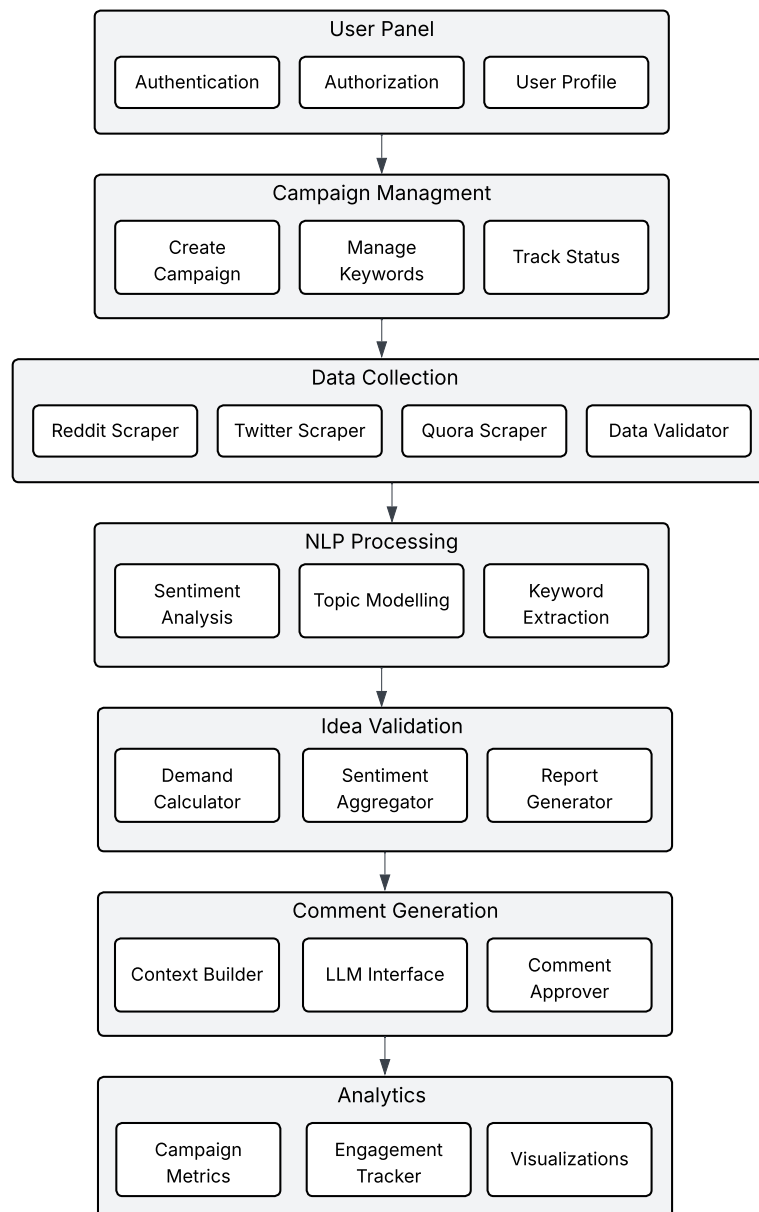


Figure 5.2: Subsystem Architecture of SurgeAI - Client

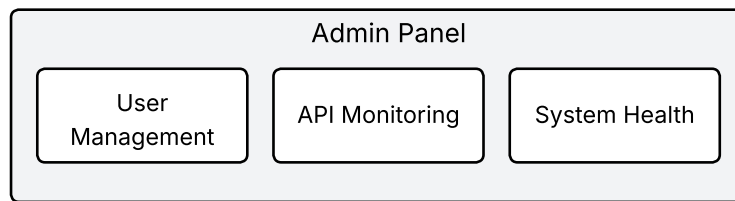


Figure 5.3: Subsystem Architecture of SurgeAI - Admin

In the above two architectures, components are divided into further subsystems. The purpose of this diagram was to further show the compositions; the working essentially remains the same as explained in the system architecture.

5.4 Architectural Strategies

It is safe to say that building any complex system like SurgeAI is just like creating a building. Before the actual construction, in our case development, we need a definite blueprint which is also referred to as the architecture. The architecture essentially tells how all the subsystems will work together to make the whole system work. It also tells us the relationship between every subsystems and that how they communicate to achieve a common goal. The architectural design of SurgeAI is based on three main goals. Firstly, it had to be scalable because with growing number of users our app should not become slow. Secondly it had to be responsiveness and this means that the user should not be blocked during time-intensive tasks like scraping and NLP analysis. Infact the user should be able to seamlessly interact with the app while the time taking services are running as background tasks. Third and last goal is maintainability according to which, adding new features and fixing errors in the application should be easy.

5.4.1 Layered Architecture for Separation of Concerns

As previously mentioned the system uses a five-layer architecture and the primary reason for this is to separate concerns. By using this approach it is significantly easy to develop and test independently. Every service and be developed and tested independently. In other words the system has low coupling which is very good.

5.4.2 Asynchronous Task Processing with a Distributed Task Queue

The combination of Redis and Celery has been used overhere and this combination handles all long running time-intensive tasks such as scraping and NLP analysis, asynchronously. This is very powerful

because this way the user will not be blocked until the long running tasks end. The user interface remains responsive and later if the system load increases, more celery workers would be created to tackle it.

5.4.3 Python-Powered Backend for AI and Data-Intensive Operations

The backend was chosen to be implemented in Python because the application is AI powered and involves NLP analysis. Python is the go to language for AI and Data Science which is why we chose this language. Apart from this python has built in libraries that are extensively used in the application. Moreover, FastAPI framework offers very high performance.

5.4.4 Relational Data Persistence with PostgreSQL

SurgeAI is powered by PostgreSQL in order to enforce data integrity for the system's highly structured and relational data. Due to this there is a definite consistency among the entites which leads to accurate analytics.

5.5 Domain Model/Class Diagram

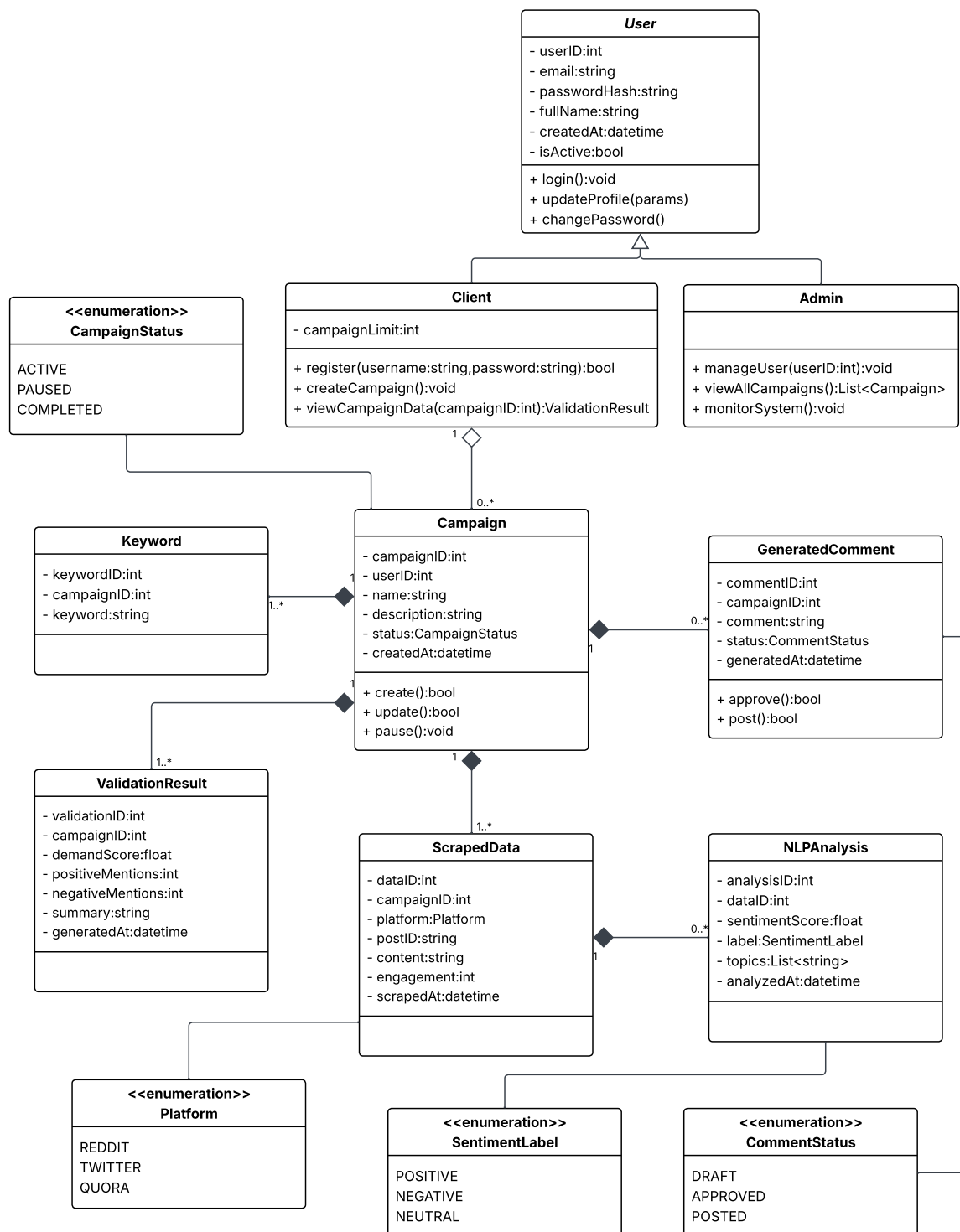


Figure 5.4: Class Diagram of SurgeAI

5.6 Policies and Tactics

Apart from architectural efficiency, we have also greatly focused on employing some key policies and tactics in order to ensure security, code quality and overall consistency.

5.6.1 Strict Data Validation at the API Boundary

Pydantic is used to ensure strict data formats for all requests and responses from API. It is a python library that is used for data validation. Data that will flow through the system will have a definite structure to it, which reduces any potential for errors.

5.6.2 Object-Relational Mapping (ORM) for Database Interaction

In order to interact with the PostgreSQL database, SQLAlchemy is used as the Object-Relational Mapper (ORM). By using this there is no need to manually write queries to setup the database. SQLAlchemy provides a platform for the python backend and the database to communicate. Instead of writing query, we create python objects that represents an entity, then this object is converted to a database entity by the mechanisms provided by SQLAlchemy. This makes database logic easier to read and maintain, reduces any risk of SQL injection vulnerabilities, and in general speeds up development.

5.6.3 Environment-Based Configuration Management

All variables that are used for configuration are managed through a .env file which is loaded at runtime. This separates application code from configuration and due to this it is very easy to deploy the application in different environments (development, testing, production) without the need of any code changes.

5.7 Conclusion

In conclusion, this chapter demonstrates the high-level and low-level design of the SurgeAI system, highlighting that a well-designed architecture can ensure modularity, scalability, and maintainability. The five-layer architecture design realizes separation of obligation, effective interconnection among sub-systems and flexible scalability in the future. The use of architectural highlights such as asynchronous processing with Redis and Celery, which is a Python based backend on FastAPI, PostgreSQL to ensure data integrity resulted in improved performance and reliability. And there is strict data validation, and environment dependent configuration management to improve your code quality.

Chapter 6 Implementation and Test Cases

This chapter explains the progress of whatever we had implemented so far for the SurgeAI system. At this stage of development, our primary focus is to implement Idea Validation through Reddit data collection and then provide the user that whether his product or service or solution is in demand by people or not.

6.1 Implementation

6.1.1 Reddit Scrapping

Our first module that is completed is Reddit Data Scraper, which collects real time discussions from many subreddits based on the keywords provided by the user. This scraper is built using PRAW which is integrated with a FastAPI backend endpoint that triggers the scraping process. PostgreSQL is used to store all the posts along with their metadata. When a user submits an idea with a specific keyword, the backend processes these keywords and uses them to query relevant communities. For each post that matches the criteria, is saved into the Scraped Data table.

6.1.2 Basic NLP Base Line

We integrated a pre trained sentiment model that is Hugging Face Pipeline and store simple sentiment score in the database. In this base line we designed and implemented the data flow such as Scraped Data → Simple NLP analysis, without full demand scoring or topic modeling yet. This analysis will provide us to know that what is the tone of the post like is it negative, positive or neutral.

6.1.3 Backend Skeleton and Core Services

In backend, we set up FastAPI structure with modular architecture such as auth service, campaign manager, scraper orchestration stubs, NLP analysis stub and validation engine stub. A user registration and login which is JWT based authentication and CRUD for campaign and keyword entities is implemented. For database we implemented PostgreSQL schema as described in data dictionary. For fast processing, we set up basic Redis instance for caching for later background jobs but full queuing is completed in FYP-2.

6.1.4 Frontend works

In frontend we implemented UI for user login, creating a campaign, adding keywords, triggering Reddit scrape, and viewing scraped posts with basic sentiment. The UI is implemented using Next.js. The backend is also integrated with frontend so that it communicates with backend REST APIs and that core

entities (User, Campaign, Keyword, Scraped Data, simple NLP Analysis) and these functionalities will be working.

6.2 Conclusion

In conclusion, about we had established a base for our project. We started with Reddit scraping to get all the posts on Reddit. The posts will be fetched based on the input of keyword the user had inputted to the system. A base of NLP has been created for sentimental analysis. Moreover, UI for different parts has been coded and they will also be integrated with the backend to show some working of the project. All the other modules will be completely implemented in FYP-2 which will give the full experience of our project.

Chapter 7 Conclusion

In conclusion, the primary focus of the SurgeAI project was on defining the problem, exploring existing work, and establishing the foundational components of the system. Through detailed analysis, it has been identified that traditional advertising and idea validation methods are both costly and are inefficient for startups. The review of related systems such as Hootsuite, Brandwatch, HubSpot, and MonkeyLearn highlighted a clear gap that no existing platform provides real time idea validation combined with AI-driven targeted marketing through conversational insights.

The team successfully documented the complete project vision including project scope, explained the problem we are going to face and what is the solution and conducting a very detailed literature review, defined the Software Requirements Specification, and designed the high-level and low-level architecture using a modular and scalable five-layer system. Functional and non functional requirements are explained and prototype pictures are also shown. In architecture design, the working system was explained, Initial implementation work was completed, including Reddit scraping, baseline NLP processing, back-end skeleton setup, and basic frontend structure. These early steps verified the technical feasibility of the project that confirms that the proposed system can be built within the available time frame and skills.

Overall, the report established a strong theoretical and technical foundation for the development of SurgeAI. The research and initial implementation validate that AI-driven market analysis and intelligent outreach can significantly improve efficiency and reduce marketing costs for startups. The progress made in this phase provides a clear direction for the advanced development and integration tasks planned for FYP-2. The future plan for FYP-2 is described below.

7.1 Future Plan

One of the major parts of the upcoming development phase will involve an advanced NLP pipeline that can handle large-scale text datasets gathered from various social platforms. This includes designing and fine-tuning transformer-based models for sentiment analysis, topic extraction, intent detection, and keyword extraction. These models will be optimized for a particular domain of idea validation and marketing intelligence. The whole NLP service should be containerized for scalability and integrated with background workers through Celery to process big batches of scraped data efficiently and asynchronously.

The most important task for the next phase is to fully implement the Idea Validation Engine. This engine will pool the output of the NLP pipeline and summarize it into meaningful campaign-level insights. It will calculate demand scores, sentiment breakdowns, pain-point frequencies, and other metrics needed

to evaluate the potential of each idea. These results would be stored in the database under the Validation Result module, and API endpoints and UI screens will be developed to enable users to check the detailed validation reports for their campaigns.

The project will also extend to include the development of an AI-driven marketing comment generator. Using OpenAI or other similar LLM APIs, the system will generate tailored marketing responses with regard to particular posts that indicate user interest or pain points. Such generated comments will be stored, reviewed, and optionally auto-posted, depending on user preference. A complete performance tracking mechanism will also be implemented that enables the system to log post interactions like upvotes, replies, and views into the comment performance table. This allows businesses to measure real-world engagement and effectiveness.

The second core requirement for FYP-2 is extending scrapers beyond Reddit. Additional functionality in the form of new modules will be developed for Twitter (X) via its official API and for Quora using scraping techniques. These shall be brought together using a common Scraper Orchestrator, ensuring that all extracted content, irrespective of the platform, would be stored in the very same scraped data schema. NLP pipeline adaptation and the validation engine shall be tuned to work seamlessly across all supported platforms for consistency in analysis and insights.

Optional but of high value will be the integration with n8n automation workflows. This will provide scheduled scraping, automated triggering of NLP analysis, sending of email or dashboard notifications, and even error or failure alerts. Together with Celery background workers and Redis queues, the system will support robust automation for long-running operations like large dataset scraping and batch NLP processing. The development phase will be concluded with extensive testing, optimization, and deployment. Testing will involve unit and integration testing for authentication, scrapers, NLP models, validation logic, and comment generation services. This will also include performance testing using realistic campaign sizes to ensure the system efficiently handles the load. When stable, the entire technology stack frontend, backend, PostgreSQL database, Redis, and optionally n8n—should be deployed on selected hosting platforms like Vercel, Render, or Heroku. Final deployment documentation is prepared along with final system documentation to wrap up the project.

Bibliography

- [1] UserTesting, “UserTesting: Human Insights Platform.” Online. Available: <https://www.usertesting.com>, 2012. [Accessed: Oct. 11, 2025].
- [2] Hootsuite, “Hootsuite: Social Media Management.” Online. Available: <https://www.hootsuite.com>, 2008. [Accessed: Oct. 11, 2025].
- [3] HubSpot, “HubSpot Marketing Automation.” Online. Available: <https://www.hubspot.com>, 2006. [Accessed: Oct. 11, 2025].
- [4] Brandwatch, “Consumer Intelligence and Social Media Analytics Platform.” Online. Available: <https://www.brandwatch.com>, 2007. [Accessed: Oct. 11, 2025].
- [5] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in *Proc. North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pp. 4171–4186, 2019. [Accessed: Oct. 07, 2025].
- [6] Meltwater, “Meltwater: Media Monitoring & Analytics.” Online. Available: <https://www.meltwater.com>, 2001. [Accessed: Oct. 11, 2025].
- [7] Buffer, “Buffer: Social Media Toolkit.” Online. Available: <https://buffer.com>, 2010. [Accessed: Oct. 11, 2025].
- [8] SEMrush, “SEMrush: SEO and Keyword Research Tool.” Online. Available: <https://www.semrush.com>, 2008. [Accessed: Oct. 11, 2025].
- [9] MonkeyLearn, “MonkeyLearn: AI Text Analysis.” Online. Available: <https://www.monkeylearn.com>, 2014. [Accessed: Oct. 11, 2025].
- [10] Talkwalker, “Talkwalker: Social Listening and Analytics Platform.” Online. Available: <https://www.talkwalker.com>, 2009. [Accessed: Oct. 11, 2025].